DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: (II)

Problem Set 1

(Sufficiency, Minimal Sufficiency, Unbiasedness)

[1] Using Neyman-Fisher Factorization Theorem find a sufficient statistic based on a random sample $X_1, X_2, ...X_n$ from each of the following distributions:

(a) $f_\alpha(x) = \begin{cases} \frac{1}{\alpha}\exp\left(-\frac{x}{\alpha}\right) & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$

(b) $f_\beta(x) = \begin{cases} \exp(-(x-\beta)) & \text{if } x > \beta \\ 0 & \text{otherwise} \end{cases}$

(c) $f_{\alpha,\beta}(x) = \begin{cases} \frac{1}{\alpha}\exp\left(-\frac{(x-\beta)}{\alpha}\right) & \text{if } x > \beta \\ 0 & \text{otherwise} \end{cases}$

(d) $f_{\mu,\sigma}(x) = \begin{cases} \frac{1}{x\sigma\sqrt{2\pi}}\exp\left(-\frac{(\log x-\mu)^2}{2\sigma^2}\right) & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$

(e) $f_\theta(x) = \begin{cases} \frac{1}{\theta} & -\theta/2 \le x \le \theta/2 \\ 0 & \text{otherwise} \end{cases}$

[2] Let $X_1, X_2, \ldots, X_n$ be a random sample from a discrete distribution with probability mass function:

$$f_\theta(x) = \frac{c(\theta)}{x^2}, \quad x = \theta+1, \theta+2, \ldots$$

where $\theta \in \{0, 1, 2, \ldots\}$ and $c(\theta)$ is a normalizing constant. Show that $T(\underline{X}) = X_{(1)}$ is a sufficient statistic for θ.

[3] Let $X_1, ..., X_n$ be a random sample with densities:

$$f_{X_i}(x) = \begin{cases} \exp(i\theta - x) & \text{if } x \ge i\theta \\ 0 & \text{otherwise} \end{cases}$$

Using Neyman-Fisher Factorization Theorem find a sufficient statistic for θ.

[4] $X_1, X_2, ...X_n$ is a random sample from $Beta(\alpha, \beta)$ distribution $(\alpha > 0, \beta > 0)$ with p.d.f.:

$$f(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Show that:

(a) $\prod_{i=1}^n X_i$ is sufficient for α if β is a given constant.

(b) $\prod_{i=1}^n (1 - X_i)$ is sufficient for β if α is a given constant.

(c) $(\prod_{i=1}^n X_i, \prod_{i=1}^n (1 - X_i))$ is jointly sufficient for (α, β) if both parameters are unknown.

[5] Let T and T^* be two statistics such that $T = \psi(T^*)$. Prove or disprove that if T is sufficient, then T^* is also sufficient.

[6] Find a minimal sufficient statistic based on a random sample $X_1, X_2, ...X_n$ in each of the following cases:

(a) $f_\alpha(x) = \begin{cases} \frac{1}{\alpha}\exp\left(-\frac{x}{\alpha}\right) & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$

(b) $f_\beta(x) = \begin{cases} \exp(-(x-\beta)) & \text{if } x > \beta \\ 0 & \text{otherwise} \end{cases}$

(c) $f_{\alpha,\beta}(x) = \begin{cases} \frac{1}{\alpha} \exp\left(-\frac{(x-\beta)}{\alpha}\right) & \text{if } x > \beta \\ 0 & \text{otherwise} \end{cases}$

(d) $f_{\mu,\sigma}(x) = \begin{cases} \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x-\mu)^2}{2\sigma^2}\right) & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$

(e) $f_\theta(x) = \begin{cases} \frac{1}{\theta} & \text{if } -\theta/2 \le x \le \theta/2 \\ 0 & \text{otherwise} \end{cases}$

(f) $f(x) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

[7] Prove or disprove that if T is an unbiased estimator of θ, then T^2 is an unbiased estimator of θ^2.

[8] (a) Let $X_1, \dots, X_n$ be a random sample from $N(\theta, \theta)$ where $\theta \in (0, \infty)$. Show that $T(\underline{X}) = \sum_{i=1}^n X_i^2$ is a sufficient statistic for θ.

(b) Let $T(\underline{X})$ be a sufficient statistic for θ. Define $U(\underline{X}) = 15T^2(\underline{X}) + 8T(\underline{X}) - 23$. Prove that $U(\underline{X})$ is sufficient for θ.

[9] Let $X_1, X_2, ...X_n$ be a random sample from $U(0, \theta)$; $\theta > 0$. Show that $\frac{n+1}{n} X_{(n)}$ and $2\overline{X}$ are both unbiased estimators of θ.

[10] Let $X_1, X_2, ...X_n$ be a random sample from $N(\theta, \theta^2)$, $\theta > 0$. Show that $\frac{(\sum_{i=1}^n X_i)^2}{n(n+1)}$ and $\frac{\sum_{i=1}^n X_i^2}{2n}$ are both unbiased estimators of θ^2.

[11] Let $X_1, X_2, ...X_n$ be a random sample from $P(\theta)$; $\theta > 0$. Find an unbiased estimator of $\theta e^{-2\theta}$.

[12] Let $X_1, X_2, ...X_n$ be a random sample from $B(1, \theta); 0 \le \theta \le 1$.

(a) Show that the estimator $T(\underline{X}) = \frac{\frac{1}{2}\sqrt{n} + \sum_{i=1}^n X_i}{n+\sqrt{n}}$ is not unbiased for θ.

(b) Show that $\lim_{n\to\infty} E(T(X)) = \theta$.

(An estimator satisfying the condition in (b) is said to be unbiased in the limit).

[13] Let $X_1, ..., X_n$ be a random sample from a continuous distribution with p.d.f. $f(x, \theta)$ which satisfies $f(\theta - z) = f(\theta + z)$ for all z. Let $T = T(X_1, ..., X_n)$ denote a statistic which satisfies:

$$T(x_1 + h, ..., x_n + h) = T(x_1, ..., x_n) + h \quad \text{for all } h$$

and

$$T(-x_1, ..., -x_n) = -T(x_1, ..., x_n)$$

Prove that T is an unbiased estimator of θ provided the expectation exists.